PROOF METHOD DECISION SHEET

A technique is useful only if it proves the part of the question that remains unfinished. This is a learning reference, not a graded test.

Before choosing a method

GIVEN: What objects and restrictions are allowed?
TARGET: What exact final statement must be established?
PROGRESS: Which claims have been justified already?
GAP: What still needs proof, or what would a counterexample look like?

Choose a useful move

Method	When to try and what to finish
Read the task	Domain, constraints, final claim and counterexample.
Similarity	Write the vertex map, then prove matching angles or ratios.
Necessity & sufficiency	Find possible answers, then verify all original conditions.
Squares & factorisation	Compute a difference, justify every factor sign.
Induction	Base case, exact recurrence, correct next step.
Modular arithmetic	List residues exhaustively; check every input class.
Bound & construction	A universal bound plus an example attaining it.
Pigeonhole	Identify boxes, objects, and what sharing a box forces.
Invariants	Show one quantity is preserved by every legal move.
AM-GM	Check positivity, equality and the direction of the bound.

Five checks before calling it a proof

- Every logical step follows from a stated fact.
- All allowed inputs, signs, cases and indices were covered.
- A maximum/minimum has both the bound and attainment.
- A classification has necessity and sufficiency, or proven equivalence.
- The last sentence directly answers the original question.